

NAZIA TABASSUM TOMA

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EDUCATION

Master of Engineering, Computer Science

UNIVERSITY OF CINCINNATI

- CGPA: 4.00/4.00

Cincinnati, Ohio, USA

August 2023 - May 2025

Bachelor of Science, Computer Science and Engineering

NORTH SOUTH UNIVERSITY

- CGPA: 3.71/4.00

Dhaka, Bangladesh

May 2017 - December 2021

RESEARCH EXPERIENCE

Capstone Research Project, University of Cincinnati

ADVISOR: [DR. NITIN CHANDERWAL](#)

Cincinnati, Ohio, USA

December 2024 - June 2026

- Proposed a Bayesian user modeling framework using targeted questions to classify users and adapt LLM outputs based on conversational style.
- Integrated GPT-based prompting to estimate behavior-driven likelihoods and update user profiles dynamically across dialogue turns.
- Developed a personalized content generation pipeline conditioned on evolving user intent without additional prompt tuning.
- Built and tested a chatbot prototype to validate real-time profile adaptation and response personalization.
- Built a Computer Science Note Generator by fine-tuning CodeLLaMA-7B-Instruct with LoRA to produce structured markdown-style explanations from topic prompts.
- Developed an MCQ Generator using a RAG pipeline with LangChain and Mixtral-8x7B-Instruct to create DSA quizzes with questions, answers, and distractors.
- Used these systems as testbeds to evaluate the impact of user profile-based personalization on output clarity, accuracy, and relevance.

Undergraduate Researcher, North South University

ADVISOR: [DR. M. ARIFUR RAHMAN](#)

Dhaka, Bangladesh

May 2021 - October 2021

- Co-authored a paper introducing a recommendation system using NLP and machine learning to recommend movies to book lovers (and vice versa) based on genre similarities such as action and science fiction.
- Collaborated in creating and annotating a combined dataset of movie and book descriptions, applying TF-IDF vectorization to analyze content similarities.
- Implemented and tested machine learning algorithms, including K-means clustering, hierarchical clustering, and cosine similarity, using subjective judgment to evaluate results due to the absence of ground truth labels.
- Explored potential applications for this system in platforms like Netflix and Amazon Kindle Store to enhance user engagement through cross-content recommendations.

Undergraduate Thesis, North South University

ADVISOR: [MR. ZUNAYEED BIN ZAHIR](#)

Dhaka, Bangladesh

January 2021 - September 2021

- Our research proposed methods to make stroke detection techniques less formidable and compares machine learning models that can predict facial drooping and vocal paralysis.
- Successful test runs were performed on facial images and voice recordings datasets and obtained 93.5% and 82.3% accuracy on facial voice datasets, respectively.
- Different models, such as VGG-16, InceptionV3, ResNet50, ANN, CNN, and LSTM, were used to compare the accuracy and find the best-performed model.
- An application was proposed designed using the FAST method and machine learning models to detect stroke patients quickly.

PUBLICATIONS	Nawar, A., Toma, N. T., Al Mamun, S., Kaiser, M. S., Mahmud, M., & Rahman, M. A. (2021, October). <i>Cross-content recommendation between movie and book using machine learning</i> . In 2021 IEEE 15th international conference on application of information and communication technologies (AICT) (pp. 1-6). IEEE.	
	Toma, N. T., & Nitin. (Accepted, 2026). BG-LLM: Bayesian-Guided LLM Framework for Personalized Learning. [16th IEEE Integrated STEM Education Conference]. <i>In Press</i> .	
WORK EXPERIENCE	Taco Tango Food Truck Ordering App	Oshawa, Canada
	SOFTWARE DEVELOPER	April 2024 - June 2026
	- Developed a custom standalone desktop app for a food truck owner to take customer orders. - Built using Python, Tkinter for the backend and frontend, and SQLite for local data management without needing an internet connection.	
	Barikoi Technologies Ltd.	Dhaka, Bangladesh
TEACHING EXPERIENCE	MACHINE LEARNING ENGINEER INTERN	July 2021 - September 2021
	- Conducted applied NLP research to parse unstructured Bangladeshi address data. - Developed spaCy-based models for geographic entity extraction. - Improved model accuracy by 15% through data cleaning, augmentation, and structured preprocessing; documented the pipeline for reproducibility and integration.	
	North South University	Dhaka, Bangladesh
	TEACHING ASSISTANT	October 2021 - May 2022
	- Delivered tutorial sessions and prepared learning materials for over 200 students. - Graded assignments, quizzes, and exams based on faculty guidelines.	
	Private Tutoring - Science and Computer Studies	Dhaka, Bangladesh
	PRIVATE TUTOR	February 2017 - January 2022
	- Provided specialized instruction in Physics, Chemistry, and Mathematics to 12 high school students, leading to an average achievement of 87% on the Secondary School Certificate examination. - Mentored a university student in core computer science subjects, including database management, C++, and Java, fostering significant academic progress over 1.5 years.	
	Introduction to GitHub Workshop - North South University	Dhaka, Bangladesh
	INSTRUCTOR	November 2020
	- Instructed 26 students on core Git functionalities, including push, pull, commit, branching, and merge conflict resolution. - Introduced fundamental project management tools within GitHub, such as Kanban boards, projects, and issues.	
	Introduction to Robotics Workshop - North South University	Dhaka, Bangladesh
	INSTRUCTOR	February 2019
	Designed and led a hands-on session for 15 students on constructing and programming soccer robots, focusing on motor control, speed, and rotation.	
AWARDS AND HONORS	Magna Cum Laude for Academic Excellence , North South University	2023
	Recognition of Service to ACM , ACM HQ, USA	2020
	Best Industrial Solution , HackNSU Season 2 Hackathon	2020
	Champion -Robotics Trivia Challenge , North South University	2018

PROJECTS	Efficient Meal and Grocery Planner (Multi-Agent System) OpenAI, LangChain, Judgeval SDK, Python		
	UNIVERSITY OF CINCINNATI	June 2025 - August 2025	
	Automated Bug Classification and Resolution Prediction System SVM, Multinomial Naive Bayes, Random Forest, BERT		
	UNIVERSITY OF CINCINNATI	August 2023 - December 2024	
	Game Moment Classification from Sports Video Using Neural Network Python, CNN		
	NORTH SOUTH UNIVERSITY	May 2021 - September 2021	
	Coupon Searching Platform Django, SQL, Web Scraping, HTML, CSS, Bootstrap		
	NORTH SOUTH UNIVERSITY	September 2020 - January 2021	
SKILLS	Programming: Python, Javascript, HTML, CSS		
	Libraries & Frameworks: Pandas, NumPy, spaCy, TensorFlow, PyTorch, Bootstrap, Django, Tkinter		
	AI: Hugging Face, OpenAI API, LangChain, RAG, LoRA, Prompt Engineering		
	ML Algorithms: Decision Trees, SVM, K-Nearest Neighbors, Logistic Regression		
	Databases & Cloud: MySQL, SQLite; AWS (EC2, S3, RDS, Route 53, CloudFormation)		
SERVICE & LEADERSHIP	Reviewer, AIED’26		
	Chair, NSU ACM-W Student Chapter, North South University.	2019 - 2020	
	Vice-Chair, NSU ACM-W Student Chapter, North South University.	2018 - 2019	